**Title**: A Statistical Method Analyzing LED Lumen Depreciation and Projecting LED Life

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**Abstract**: There is a strong need for a method to represent the potential life of LED products as a critical part of design decisions including cost-effectiveness analysis. The IES LM-80 test method is in place to collect lumen depreciation data but it does not provide for estimation of future long term depreciation. Separate estimation methods are being considered (TM-21) and this paper describes the analysis of a series of LED degradation models and subsequent development of an estimation method. The work involves analysis of a set of engineering models to determine their applicability and define a structure for their use in LED lumen output life based on a lumen output level such as the $L_{70}$ metric. The analysis has provided valuable information on methods effectively estimating LED life time, and impacts of measurement uncertainties, test duration, interval and other test conditions on selecting degradation models and LED life time projection. A set of guidelines are recommended to estimate LED life from data obtained using the current LM-80 test method.

**Key words**: LED, degradation model, LM-80, statistical analysis, $L_{70}$ projection, TM-21

## 1. Background to the investigation

Solid state Light Emitting Diode (LED) technology is the new lighting technology and is being promoted as a technology with many beneficial attributes. One of these attributes is the potential for extremely long life far beyond other lighting technologies. The life of typical non-LED lighting (lamps) is historically based on the time when a percentage (typically 50%) of a sample of lamps experience filament, cathode, or similar failure and ceases to produce light). This failure based rating system is sensible for most technologies because lamp filaments or cathodes fail before significant lumen depreciation is experienced.

In contrast, LED technology does not require filament or cathode type components and its other components are fairly robust and resistant to breakage or failure. However, the LED product will continue to experience lumen depreciation and may reach a point where the light output is below a level sufficient for its intended use. For design and operation purposes, this point is what the industry is considering to be a useful measure of the useful life of an LED chip, module or array of LED chips. Initial work by the LRC(full name) identified a 70% lumen output value ($L_{70}$) as a useful metric for useful life of LED products in typical applications. Other LED lighting system components can fail or break and these must also be considered in a true metric of LED product life, but the measure of lumen depreciation to a useful point can be a primary element of a useful LED life metric.

A primary issue is that the emerging LED industry does not have a uniform method of predicting or estimating future LED life such that the estimates made can cover a wide range of timeframes (50,000 hours to infinity). Specific attributes of LED technology make it difficult to accurately measure their true lumen depreciation in an operating environment and this contributes to the wide variety of life estimates. The first of these attributes is the potential long life of the product. It is considered impractical to do complete life tests on a product that may last 40,000 to 100,000 hours or more (4.5 to over 11 years of continuous testing). A second issue is the sensitivity of LEDs to heat more so than most other lighting technologies. This means that true lumen output testing for LEDs needs to be done on an absolute basis as a complete luminaire system to capture the true operating conditions of the complete product. Relative

testing (lamp separate from fixture) or accelerated testing at higher temperature than the end applications is not considered appropriate for LED products.

As with all lighting products, the industry needs a useful metric for lighting product life in order to make good design and application decisions. With the specific LED attributes in mind, the industry has already developed a test method for collecting accurate and repeatable lumen depreciation measurements of LEDs known as LM-80-2008. LM-80 describes the test conditions and procedures for collecting depreciation data with a minimum collection period of 6,000 hours at 1,000 hour intervals and at three specified and chosen temperature conditions. This is critical in order to provide varying temperature depreciation curves to match the actual in-situ temperature of the LED chip, module or array when it is installed in its luminaire (which affects its operating temperature). However, LM-80 does not provide any method for using these data to estimate a useful life beyond the 6,000 hour test period. As a follow-on to LM-80, an additional IES document, TM-21 was started to provide a method of life estimation based on the data collected according to LM-80. TM-21 intends to provide a conservative and fair method of estimating useful lumen output life (i.e. $L_{70}$) such that the industry can effectively compare potential useful life of various products.

Initial work by the LRC, identified similarities between then known LED depreciation data and a standard exponential decay curve. This relationship was used early in programs such as Energy Star to develop estimated of LED life. However, since then it has become clear that LED products do not all decay according to this mathematical format. Figure 1 compiles the data from spot photometry of long-term testing samples in the DOE CALIPER program. With the rapid development and deployment of LED technology in various consumer products, there is an urgent need to indentify appropriate engineering-based models to effectively describe decay structures of common LED products and to develop estimation methods for projecting LED life. Further work by Emil Radkov identified a set of multiple models that could describe decay structures of various LED product components and potentially some depreciation phenomena of LED products as a whole. Based on these models, researchers at Pacific Northwest National Laboratory (PNNL) conducted statistical analysis using theoretical and real LED life data obtained following the LM-80 test protocols. The results of this statistical study were used to identify a proposed a set of rules for selecting the best fit model for a LED product and an estimation method for projecting a useful LED life ($L_{70}$). Suggestions were also made to improve the current LM-80 test method for more accurate LED life estimation.

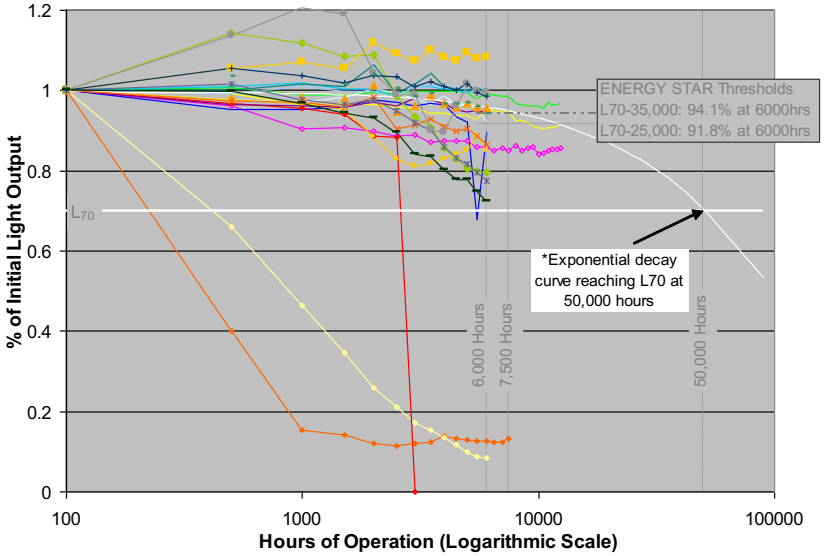



Figure 1 Spot measured luminance depreciation after long-term continuous operation

## 2. Lumen Depreciation Analysis Methods

Degradation of LED chips, encapsulants, reflectors and other components impact LED life in a variety of ways depending on the physical mechanism of each degradation phenomenon. Table 1 describes a set of mathematical models and their combinations originally identified by Radkov that are used to describe the physical degradation phenomena of LED components and products. In the absence of other LED lumen depreciation estimation methods, our statistical analysis used this set of models for exploration and study toward a structured industry acceptable LED life estimation method.

| # | Parameters | | | Decay Rate | Closed Form Solution | Degradation Mechanism |
|---|---|---|---|---|---|---|
| | $k_3$ | $k_2$ | $k_1$ | | | |
| 0 | 0 | 0 | 0 | $dI_v/dt = 0$ | $I_v = I_v^0$ | None observed |
| 1 | 0 | 0 | 1 | $dI_v/dt = k_1$ | $I_v = I_v^0 + k_1(t-t^0)$ | Drift of charged defects in chip |
| 2 | 0 | 1 | 0 | $dI_v/dt = k_2 I_v$ | $I_v = I_v^0 \exp[(k_2(t-t^0)]$ | Thermal decomposition of encapsulant |
| 3 | 0 | 1 | 1 | $dI_v/dt = k_1 + k_2 I_v$ | $I_v = (I_v^0 + k_1/k_2) \exp[k_2(t-t^0)] - k_1/k_2$ | Model 1+ Model 2 |
| 4 | 1 | 0 | 0 | $dI_v/dt = k_3/t$ | $I_v = I_v^0 + k_3 \ln(t/t^0)$ | Corrosion of reflector |
| 5 | 1 | 0 | 1 | $dI_v/dt = k_1 + k_3/t$ | $I_v = I_v^0 + k_1(t-t^0) + k_3 \ln(t/t^0)$ | Model 1+Model 4 |

Table 1 Engineering-based models for describing degradation mechanisms of various LED components and products

It needs to be understood that the application of any set of models to represent future LED degradation and life projection is considered promising but not proven. These models are unlikely to cover all LED decay profiles and it is not known at this stage which model may describe behaviors of a particular product better than others. Parameters such as coefficients of determination ($R^2$) and the residual sum of squares (SSE) are generally used to gauge the goodness of fit of models for any given data set. Preliminary results indicate for most LED decay data the differences of $R^2$s and SSEs among several models are not significant enough to determine the absolute best model; therefore, the accuracy of future projections for life is very low. Furthermore, uncertainty (data noise) is inherent to measurement data due to apparatus repeatability, time dependent factors, operation consistency, etc. Our initial statistical analysis on effects of data noise suggests that even a small amount of data noise could result in poor model differentiation and significant life projection error as a consequence. Therefore, to gain insight into the various mathematical models for LED degradation, life projection and effects of various test related parameters, a statistical analysis approach appears necessary and was chosen for this study.

In our statistical analysis, five hundred decay scenarios were generated by adding noise to the measurements to represent uncertainty to each decay data randomly. Model 1 to 5 in Table 1 were coded in R, a statistical software package, fit to each potential decay scenario and projected into the future. Evaluations were made on the set of simulations based on analyzing the statistics of SSE and $L_{70}$ projection values stored for each simulation.

## 3. Data

We first used constructed decay data which emulated several typical LED decay scenarios, as shown in Figure 2. The candy cane scenario exhibits a hump in the initial data that characterizes a transient effect in

LED lumen output during the "warming up" period. Linear and curved linear data are two common LED decay profiles before other decay trends manifest themselves. Accelerated decay is also observed with some LED products in which the lumen output drops quickly. In the limited test period, lumen output of some LED products may not show any sign of lumen depreciation, some lumen output may flatten out after decreasing for a while, and some may rise again above the earlier lumen output after initial decrease. These are captured in the flat, asymptotic and U-shaped scenarios respectively. 1% measurement uncertainty recommended by the NIST LED standard testing lab was used in simulation as the default. In order to make this simulation result applicable to real data taken using the LM-80 test method, constructed data were designed with a 1,000 hour interval and 6,000 hour total duration. Based on the results of these analyses, a set of model selection rules and methodologies to project LED $L_{70}$ life were suggested and are discussed in the Conclusions Section.

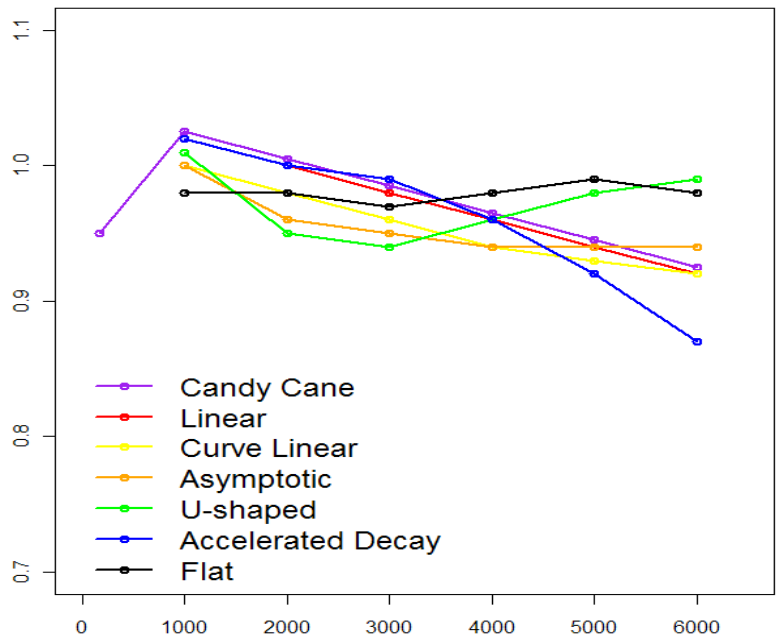


Figure 2 Constructed decay scenarios reflecting typicle LED decay profiles

To validate the model selection rule set, real LED decay data were obtained from LED manufactures with product names and operation conditions removed for unbiased analysis. These decay data were measured using the LM-80 test method, with some having additional data taken at shorter measurement intervals than 1,000 hours and/or beyond 6,000 operation hours. All data were unprocessed prior to analysis with the exception of being normalized at the first data point. Statistical analysis results of 6,000 hour data and all available data of the same product were compared in the results section.

Finally, to further illustrate effects of test related parameters such as shorter measurement intervals, longer test durations, and repeated measurements, constructed decay data with known models (linear, exponential, logarithm, combination of linear and logarithm) as shown in Figure 3 were used to simulate the effects of differing data collection requirements. Measurement intervals of 168 hours and 500 hours were simulated to compare with 1,000 hour interval. Test durations of 7,000 hours, 8,000 hours, 10,000 hours, 12,000 hours and 15,000 hours were simulated to compare with 6,000 hour duration. In addition to one data point per measurement, two and four replicates of data at the same time point were also simulated. Random noise was applied to the replicates in simulation so they truly represent multiply measurements at the same time stamp. The probabilities of correct model selected when compared against the known models were calculated in 5,000 simulations per model with 1% and 2.5% measurement

uncertainty applied respectively. The results of this analysis are discussed in the Results and Analysis Sections.

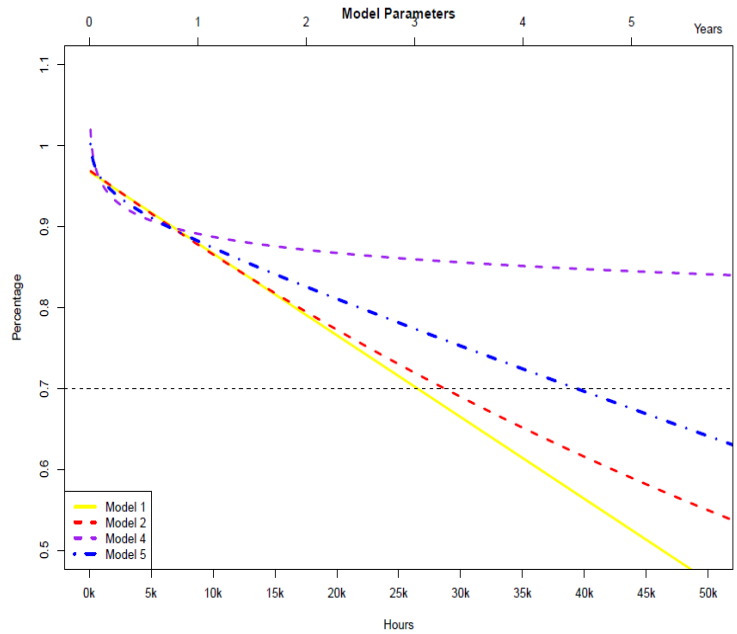
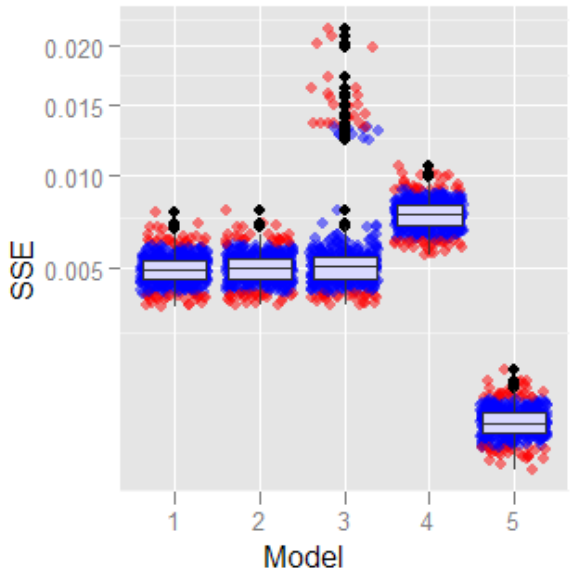



Figure 3 Data constructed based on the four known models

## 4. Depreciation Analysis Results

In the study of constructed decay data in Figure 2 and real LED decay data, the statistical distribution of SSE values presents an overall picture of the goodness of fit for each model at a given set of decay data and measurement uncertainty. Using the candy cane decay data as an example, as shown in Figure 4a, the x-axis is the model number explained in Table 1; each dot is a calculated SSE value stored in simulations. The boxes cover from the 25$^{th}$ to 75$^{th}$ percentile of the SSE values for each corresponding model. The red dots represent the top and bottom 10$^{th}$ percentiles of the SSE population for each corresponding model. The decay profiles of the different models for the same data set are plotted in Figure 4b in which the individual data points are the constructed data with 1% noise. The $L_{70}$ projection distribution for each model is expressed using a trimmed mean (thick line) and an error bar representing 25$^{th}$ to 75$^{th}$ percentile of the $L_{70}$ values on X-axis. In this example, model 5 (combination of linear and logarithm) has significantly smaller SSE than the other models, therefore, it is the obvious selection as the best fit model for this data. Though $L_{70}$ projection values can be anywhere within the distribution calculated for model 5, we suggest using the lower 25$^{th}$ percentile of the $L_{70}$ values as the final life projection for this data to avoid being overly conservative.



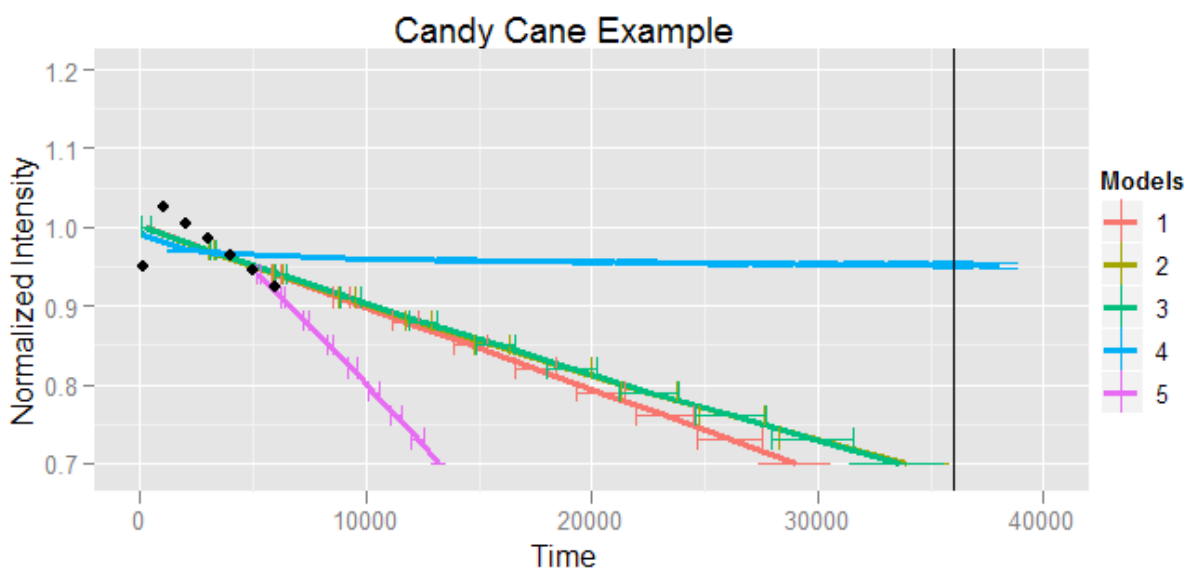
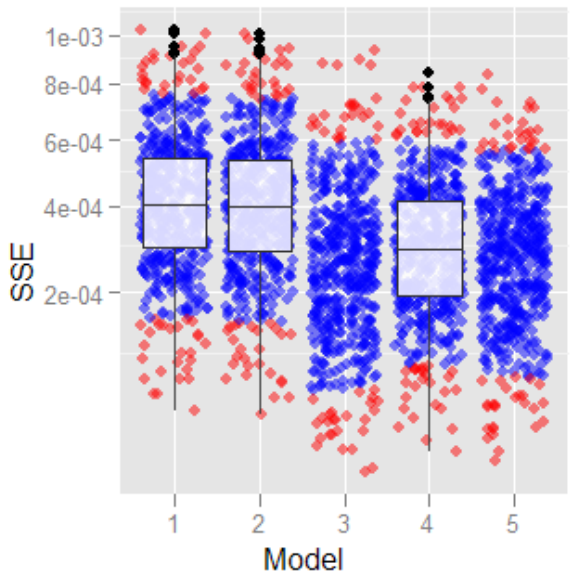




Figure 4a: Box plot of SSE values for each model in the candy cane scenario
Figure 4b: Decay profiles of each model in the candy cane scenario

In many decay scenarios, the differences of SSE values among several models may not be significant enough to pick an obvious winning model. In an example of a real set of LED decay data, shown in Figure 5a, measured at every 1,000 hours for a total of 6,000 hours using the LM-80 test method, the differences between SSE values of all models are not great enough to make a selection. In this study, we used the $10^{th}$ and $90^{th}$ percentiles of the two adjacent models as criteria to separate models. A model is selected only if its $90^{th}$ percentile is equal or lower than the $10^{th}$ percentile of the adjacent models. The $L_{70}$ projection values of all models disperse from 40,000 hour range to infinity as shown in Figure 5b. To be realistic in LED life projection and protect consumers' interest, we suggest choosing the most conservative model of all which projects the shortest $L_{70}$ life for this LED product when none of the

models stands out as the best fit. In this case, model 1 projects the shortest L70 life with the lower 25th percentiles of its distribution at approximate 38,000 hours. It needs to be noted, when more than 15% of $L_{70}$ values of one model are calculated to be infinite, the 25-75 percentile box is not drawn on the SSE plot and the model can't be selected as the best model regardless the SSE values. In this example, models 3 and 5 were excluded from model selection.



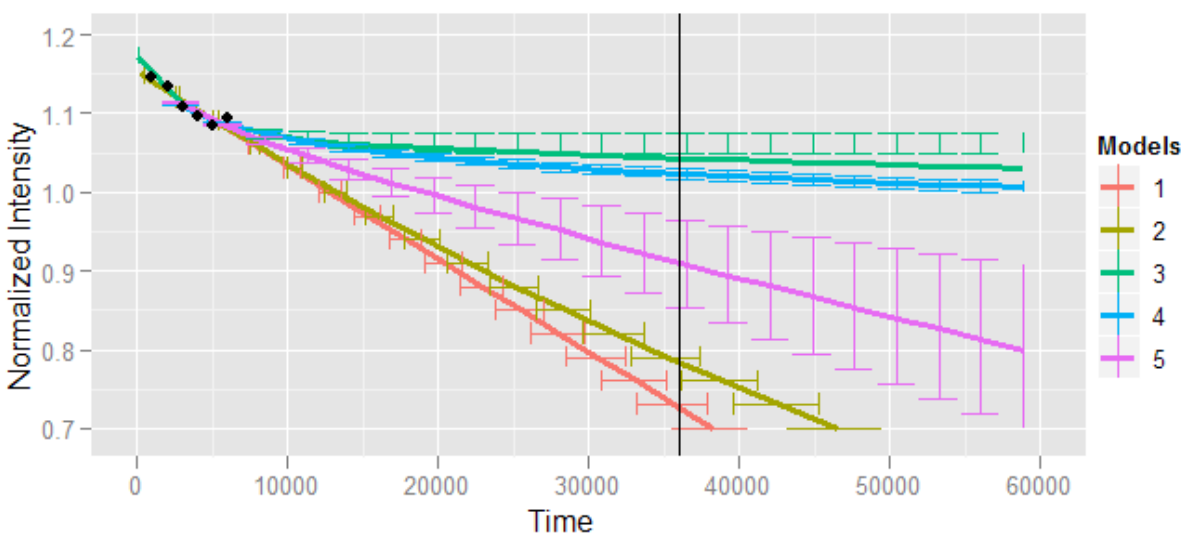




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Figure 5a and 5b Box plot of SSE and Decay profiles of each model including $L_{70}$ values using real LED decay data taken using the LM-80 test method

This LED product may have a potential life time longer than the projection of model 1. However, at the 6,000 hour test duration, there are too few data points to build an accurate model. The impact of noise on this small number of data points is also detrimental to model fitting and always warrants large errors, which translates to more disparate $L_{70}$ predictions for a given model in Figure 5b. Furthermore, the true decay profile of this particular LED may not yet be known with only 6,000 hours of data. When the same product was measured every 1,000 hours up to a total 15,000 hours, the goodness of model fit and

predictability of $L_{70}$ life were indeed improved. As shown in Figure 6a, models 4 and 5 have clearly lower SSEs than models 1 and 2, which would suggest excluding models 1 and 2 from model selection. Model 3 was excluded from model selection because 15% of calculated $L_{70}$ values were undefined, infinite, or incalculable. Because the overlap of SSE values between models 4 and 5 is greater than 10% of their populations respectively, the best model is decided upon the shorter, or more conservative, $L_{70}$ life. As a result, model 5 with the shorter $L_{70}$ life of the two is selected as the best fit model for this LED product. The $L_{70}$ life of this LED product is estimated at 130,000 hours in Figure 6b. In comparison, the results with only 6,000 hours of data show much broader $L_{70}$ distribution had estimated 38,000 hours of life. The available longer test data reduced model errors and extended the life projection for the same product.

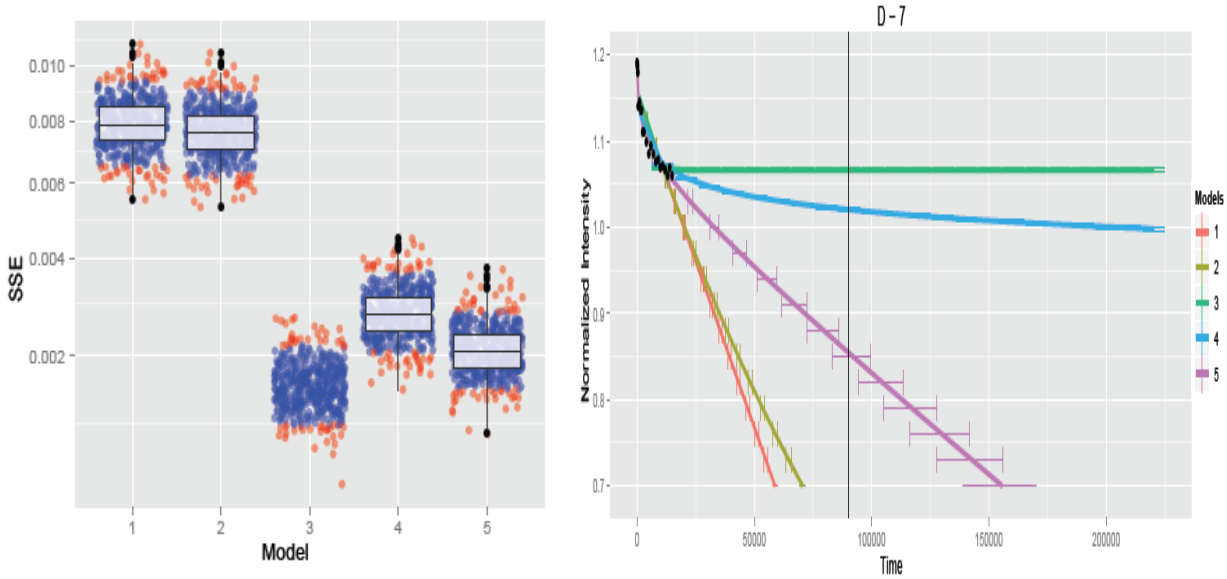


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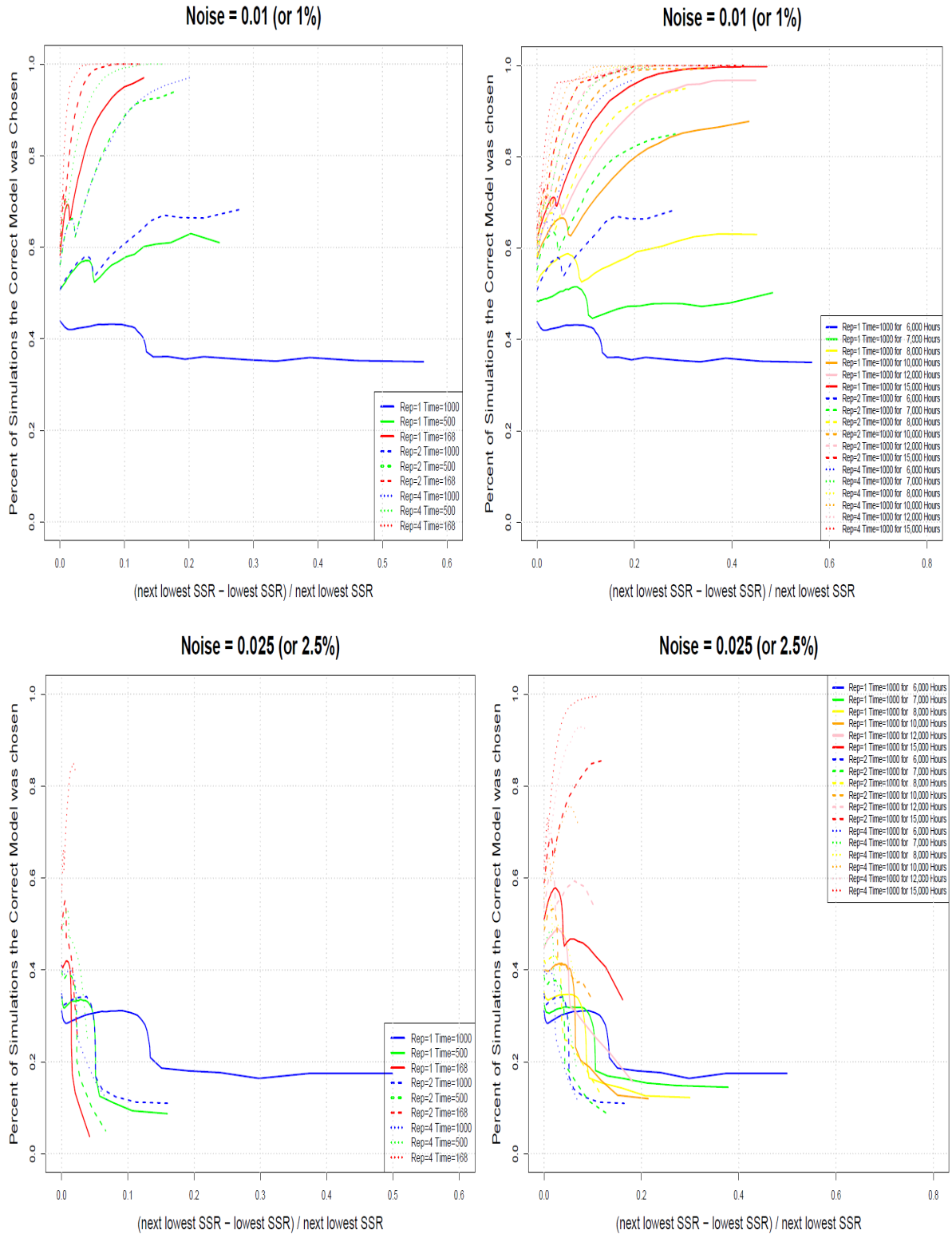
Figure 6a and 6b Statistics of SSE and $L_{70}$ values of decay data taken at longer test duration than the LM-80 test method for the same LED in Figure 5a and 5b

Simulations and statistical analyses were conducted on 28 other LED products with data at shorter measurement intervals and longer test durations than what LM-80 specifies. The sample size of these LED products ranged from 6 to 30 samples of each. At least 3 sets of data per product were selected to represent the high, low and median lumen output at the end of degradation measurement. Although degradations of different samples can exhibit different profiles, the observations of these selected (three) samples was considered representative of what would be observed across the entire sample.

The questions naturally arise regarding how long an LED sample needs to be measured in order to accurately understand model LED degradation and to project LED life accurately? What is an adequate measurement interval? Would replicates at each measurement increase modeling accuracy? Does lab measurement uncertainty matter? Although the real LED degradations are much more complex than theoretical scenarios, the study on constructed decay data with known models allows a microscopic view of effects of the above factors and suggests some clues towards answering the above questions,. The analysis results of this study are presented in Figure 7a-d, in which probabilities of a correct model selected were plotted against the difference of the lowest and next lowest SSEs. The probability of a correct model selected increases as the measurement interval decreases, the test duration increases, and/or more replicates are taken per measurement.




(change SSR to SSE, change the x-axis division on the right figure to 0.1)

Figure 7a-d Probabilities of a correct model selected by varying several test related parameters

It is clear the probability of a correct model being selected is very low using the current LM-80 test method at 1,000 hour interval and 6,000 hour duration. Furthermore, testing lab uncertainty shows a great impact on the probability of a correct model selected. Assuming a testing lab's measurement uncertainty is 1%, a LED product has a known degradation model, and the calculated lowest SSE of the model is smaller than the 90% of the next lowest SSE, Figure 7?? Shows that for the model to pick the correct model over 95% of the time, the LED need to be measured at least once every 168 hours or at least twice every 500 hours when the measurement duration is restricted to 6,000 hours. If the measurement duration can be extended to 7,000 hours, at least four repeated measurements need to be taken every 1,000 hours. As the measurement duration is extended, less repeated measurement is required, although it still requires two repeated measurements every 1,000 hours for 15,000 hours to achieve 95% probability of a correct model selected. This analysis provides guidance on indentifying the limitation of the current LM-80 test method and the directions for improvement. It is still unrealistic to propose exact test parameters for all LED products because the degradation profiles of most LED products are yet to be found at the current development stage of the industry.

## 5. Conclusions

The analysis of the various models and the many considerations of "noise", model fit (SSE), and $L_{70}$ projections provide valuable information on how to use the various models in defining a method of estimating useful ($L_{70}$) life of LED products. The summary of the analysis provided the following important model attributes and considerations for their use in life estimation:

- Model0 (Flat)-Only suitable for data that doesn't show any decay within the measurement duration but generally should not be considered as all LEDs will degrade sooner or later.
- Model 1 (Linear) - Although non-linear data is expected, within the measurement duration a curve may not yet be formed making this an appropriate model.
- Model 2 (Simple Exponential) – Historically a good fit for typical decay experience early with LED products
- Model 3 (Linear + Exponential) - This model can exhibit multiple data fitting forms and often asymptotes to a steady non degrading horizontal line which provides infinite $L_{70}$ projections that are not useful for industry use.
- Model 4 (Log) – This model also provides similar infinite $L_{70}$ projections.
- Model 5 (Linear + Log) - Tends to overfit limited data which leads to increased uncertainty in $L_{70}$ projections but is also commonly a good fit.Most models behave poorly if LED "warm-up" (data rising prior to descending decay) measurements are included.

The results of the modeling show that most models (1-5) can provide useful fit to one or more of the varieties of decay formats seen with real data testing to date. Therefore it was considered useful to evaluate LED depreciation data collected per LM-80 protocols using all models to ensure that when one was a good fit, that it was evaluated. While this could be done in a straightforward manner, the issue became choosing a model or models to represent the decay and use for the basis of a life projection. The summary results of the modeling along with general statistical rationale was used to develop a set of "rules" or procedural steps to be used in determining a life projection. Note that at the time of the writing of this paper the variable in **bold** shown in the rule set below were subject to TM-21 committee review and change.

1. A first step is to determine the appropriate set of LM-80 data to process. This will be based on the in-situ temperature of the LED product determined from a predetermined temperature measurement point with the product installed in its luminaire and operating under normal conditions.
2. A second step is to remove any "warm-up" measurements because of their general conflict with good model fitting. The rule was based on determining true warm up data that would not be considered

within typical data set noise. Therefore, the highest lumen output data point in the data set is compared with the time 0 point and if it exceeds the time 0 point by a value larger than the data set uncertainty (**1**% for example) then the data points prior to this highest point are eliminated from model fitting.

3. The third step involves fitting the remaining data to models 1, 2, 3, 4, & 5 and simulating the fit **500** times using the noise variability of the rated laboratory uncertainty (proposed **1**% as a default in the absence of a test laboratory specific value).
4. Then, calculate the SSE and $L_{70}$ values of the **500** simulations for each model.
5. At this point determine best fit models (if any) according to the following "rules":
   a) If the SSE of one model is smaller than those of the other models without overlapping them by more than **10**%, use that model (technically using a t-test as the statistical comparison of SSE values).
   b) If the SSE values of a group of models do overlap by more than **10**%, and this group's SSE values are smaller than those of the other models using the same **10**% overlap criterion, use the smallest, or most conservative, model in this group.
   c) If the SSE values of all models overlap more than **10**%, use the model with the smallest $L_{70}$ value.
6. With the winning model(s) chosen (based on step 5 above), use the lower **25$^{th}$** percentile of the $L_{70}$ values from the winning model(s) as the $L_{70}$ projection for this set of LED data.
7. And finally as a reality check to avoid unrealistic life estimates even when models are considered good fits, apply a **6** times rule. If the $L_{70}$ projection is > 6 times the measurement period, use 6 times that measurement period as the final $L_{70}$ projection.

It is understood by all concerned that this method of determining LED product life is not going to be completely accurate given the limited data and issues with model fitting in general. However, it is a considered to be a conservative and comparable method that treats all products fairly.

**Acknowledgement**

**References**